

SustainaFund — Portfolio Selection

Interim review

\$100M mid-term mandate · 1,087 stocks · MIQP in FICO Xpress

What we have, why this model, how the data was cleaned, and what the numbers say

- The model is a mixed-integer quadratic program — Markowitz with cardinality, concentration and ESG constraints, solved in FICO Xpress in under a second.
- Every non-obvious choice in it was measured rather than assumed, and the measurements are reproducible from the scripts.
- One result is negative and reported as such: out of sample the optimiser and an equal-weight benchmark are statistically indistinguishable on risk-adjusted return.

The problem

Two objectives that cannot both be maximised, under six hard constraints

Constraint	Value
Budget	\$100,000,000, fully invested
Position size	1% - 20% of budget if held
Diversification	at least 30 different stocks
Region	no region above 60%
ESG, weighted	at least 70
ESG, per stock	at least 30 (added)
Sector	no sector above 30% (added)

The last two are additions to the brief, not requirements.

The trade-off

- Maximise return — the weighted sum of expected returns. But expected return is not given in the data; it has to be estimated from ten years of prices, and the choice of estimator is itself part of the work.
- Minimise risk — the variance of the portfolio, which needs the covariance between every pair of stocks. This is what captures diversification: combining stocks that do not move together lowers total risk.
- Higher return comes with higher risk, so there is no single best portfolio — only a frontier. We present three points on it for three risk appetites.

The model

Mixed-integer quadratic program, solved with FICO Xpress in 0.3-1.1 s

<i>minimise</i>	$w' S w$	portfolio variance
<i>subject to</i>	$w' \mu \geq \text{beta}$	target return, swept
	$\sum w_i = 1$	budget fully invested
	$0.01 y_i \leq w_i \leq 0.20 y_i$	1-20% if selected
	$y_i \in \{0, 1\}$	selected or not
	$\sum y_i \geq 30$	minimum holdings
	$\sum w_i \leq 0.60$ per region	
	$\sum w_i \leq 0.30$ per sector	
	$\sum \text{ESG}_i w_i \geq 70$	weighted ESG
	$\text{ESG}_i \geq 30$ if held	per-stock floor

S = covariance matrix, μ = expected returns

Binary variables are required, not a workaround: Xpress semi-continuous variables express "0 or in [1%, 20%]" natively but cannot be counted, so the 30-stock constraint still needs them. Tested - dropping the link returns 26 positions while reporting 30.

Why quadratic and not linear

- The brief offers two routes to risk: per-stock volatilities, which give a linear model, or the covariance matrix, which gives a quadratic one. We built both.
- A linear objective is the risk you would bear if every pair of stocks moved together. It is blind to diversification: swapping a stock for another of equal volatility never changes it.
- Measured: at the same expected return the linear model carries 19.2% more true risk, and it always holds exactly the 30-stock minimum - adding a 31st cannot improve its objective.
- For its own portfolio the linear objective reads 21.95% while that portfolio's actual volatility is 13.42%. The 1.64x gap is the diversification it cannot see.

How the data was cleaned

Seven decisions, each with the number that justifies it

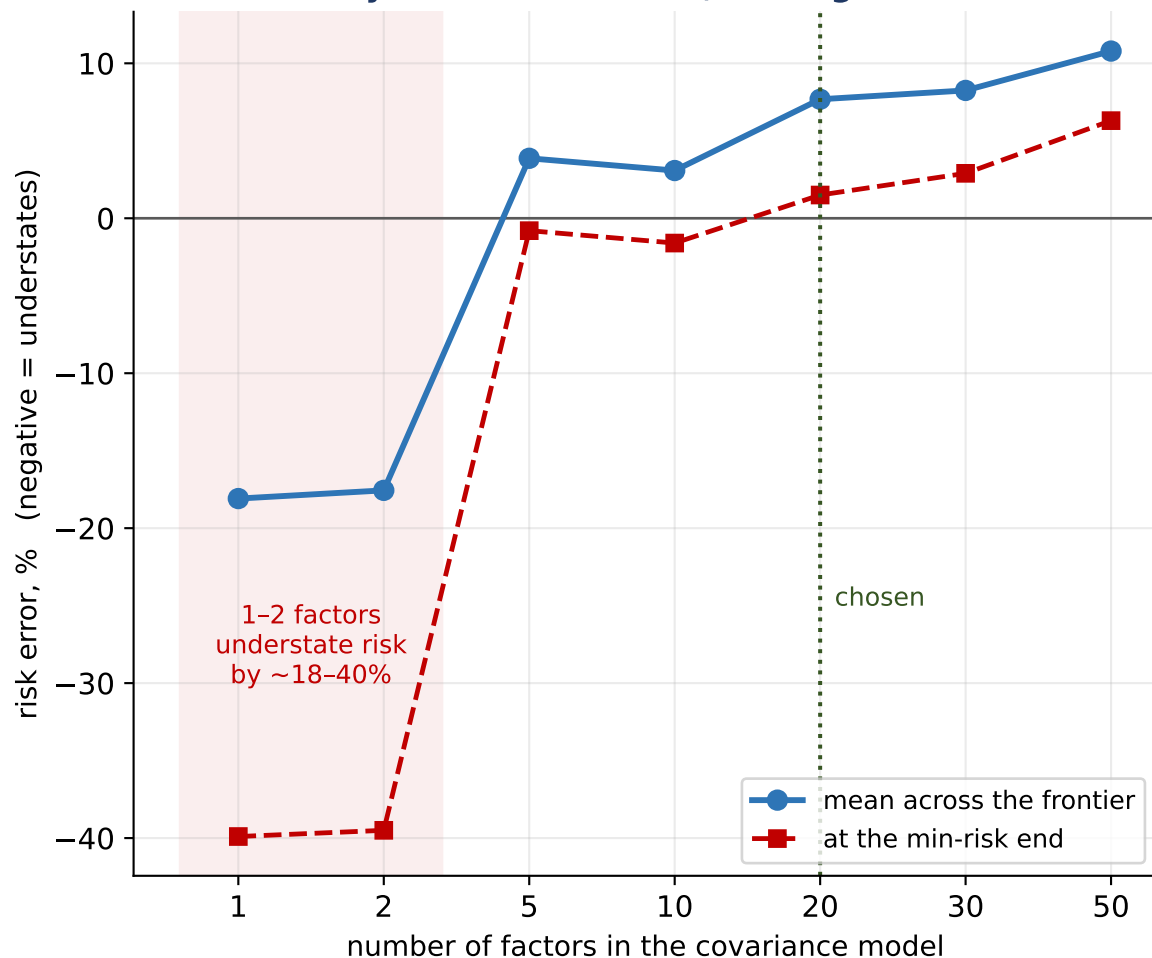
Step	Decision	Why - the measured reason
Missing ESG	49 of 1,093 filled with the 25th percentile within the region	the 25th percentile, not the mean, so a missing score is not rewarded
Price gaps	forward-filled, never before a stock's first trading day	every internal gap is exactly 1 trading day, checked on consecutive runs
Dividends	switched to total-return prices	median expected return 7.80% → 10.39%; the rise is ordered by dividend yield across sectors
Currencies	converted to USD at daily ECB rates	a local-currency Σ understated the min-variance book's risk by 19.7%
Discontinuities	ZEG.L and BMPS.MI truncated, not dropped	identical closes then a large jump = corporate action; a return across it is meaningless
Outliers	flagged, not removed	a 5σ per-stock rule fires on 0.31% of observations, mostly genuine events
Expected return	James-Stein shrinkage by sample size	a raw mean gave 88% p.a. on 460 observations; after shrinkage no stock is negative

Universe 1,093 to 1,087. Six stocks are empty in the total-return source file (three of them REITs) and were dropped rather than back-filled from unadjusted prices, which would have cost them ~3pp of return for a data defect.

Why these estimator choices

Each one changed the answer, and each was measured

Why 20 factors: a cliff, not a gradient



The three that moved the result most

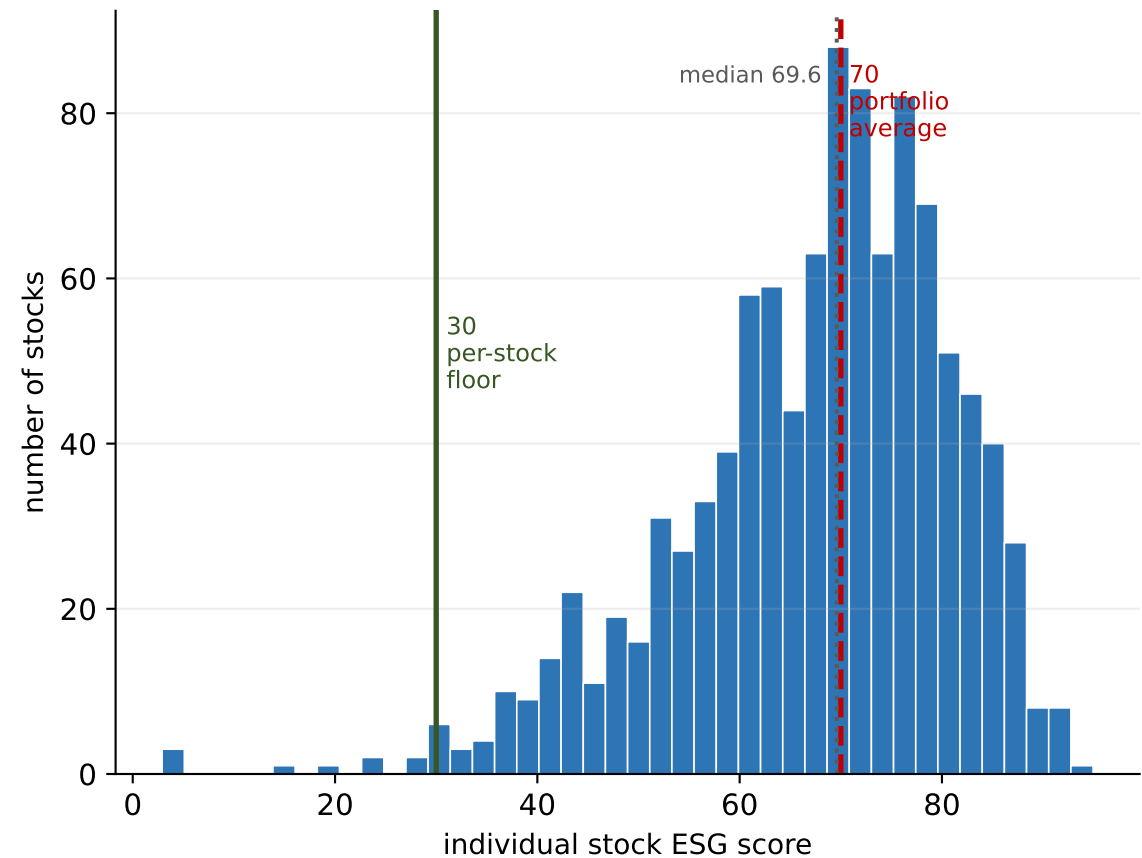
- Prices → USD. Returns are ratios, so price levels cancel — but the FX return does not: $\ln(P_{\text{usd}}) = \ln(P_{\text{local}}) + \ln(fx)$. It barely moves μ ($<0.3\text{pp}$, currencies offset) yet puts a shared factor into Σ . At the same true risk the corrected model returns 12.78% against 7.07%.
- μ shrunk. A raw historical mean is very noisy and the optimiser preferentially picks whatever that noise flattered. Shrinkage cut the frontier's top from 38.8% to 16.6% — and 38.8% would not have survived one question.
- Covariance → factor model. The complete-history rule silently excluded 96 recent listings. A factor model fits betas on whatever days a stock has, keeps all of them, and is positive semi-definite by construction.

Two things we tried that did not work, reported as such: bootstrap resampling of the weights (no effect on stability) and an explicit region factor (absorbs the structure, leaves the risk error unchanged).

Choosing the ESG floor

The per-stock floor was picked from the distribution, not guessed

1,093 stocks — where the two thresholds sit



Candidate floor	Stocks	Share
ESG < 10	3	0.3%
ESG < 20	5	0.5%
ESG < 30	10	0.9%
ESG < 40	41	3.8%
ESG < 49.3	110	10.1%
ESG < 58.6	234	21.4%

Lowest ESG	Score
TUB.BR	2.9
KBCA.BR	4.2
SATS.OQ	5.0
CVNA.N	15.2
TTD.OQ	18.7
BMEB.L	22.9

Two things this settles

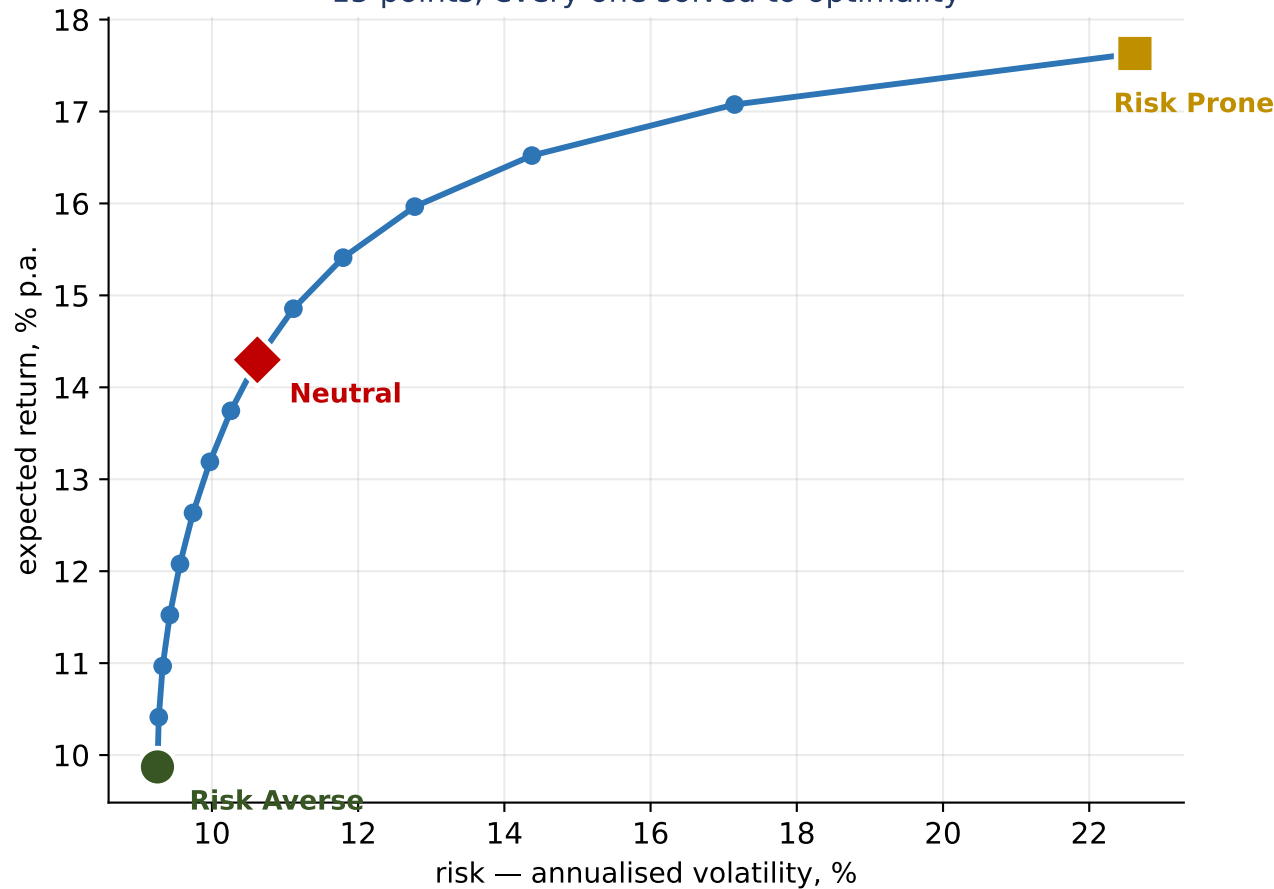
- The universe median is 69.6, essentially the 70 the brief asks for as a weighted average. So the ESG constraint means "above the median on average" — which is why it binds in every solve yet costs almost nothing.
- A weighted average of 70 can still be met while holding names scored in the twenties. The floor of 30 removes ten stocks, 0.9% of the universe, and closes that gap. Berkshire Hathaway at 24.1 is the most recognisable of them.

- 40 would have cost four times as many stocks (41) for no clear principle as 30 is the smallest floor that removes the outliers

Results — the efficient frontier

1,087 stocks; the three scenarios are the two corners and the best trade-off between them

15 points, every one solved to optimality



Scenario	Return	Risk	Ret/Risk	Held	ESG
Risk Averse	9.87%	9.26%	1.067	42	70.0
Neutral	14.30%	10.62%	1.346	30	70.0
Risk Prone	17.63%	22.62%	0.779	30	70.0

Definitions follow the brief: minimum risk, maximum return, and the highest return/risk ratio between them. No arbitrary target and no risk-aversion parameter.

The neutral book is the recommendation

- Best risk-adjusted profile at 1.35, 30 holdings, largest position 12.9%, and no position at either the 20% cap or the 1% floor.
- ESG binds at exactly 70.00 at every one of the 15 points, so the constraint is always active — and costs almost nothing.
- The frontier is steep to the left and flat to the right: past roughly 14% risk, extra risk buys very little extra return. That is the argument for not choosing the risk-prone corner.

Three investment scenarios

As the brief asks: minimise risk, maximise return, and the best trade-off between them

Risk Averse	
<i>lowest risk on the frontier</i>	
Expected return	9.87%
Risk	9.26%
Return / risk	1.067
Holdings	42
Weighted ESG	70.0
Top sector	Consumer Defensive 30%
Top country	Switzerland 47%
Largest positions	
MOBN.S	9.7%
ALLN.S	7.6%
GALE.S	6.4%
BCVN.S	6.3%
WMT.OQ	5.0%

Neutral	
<i>highest return / risk ratio</i>	
Expected return	14.30%
Risk	10.62%
Return / risk	1.346
Holdings	30
Weighted ESG	70.0
Top sector	Consumer Defensive 27%
Top country	United States 48%
Largest positions	
WMT.OQ	12.9%
GALE.S	10.2%
MOBN.S	10.1%
COST.OQ	8.2%
BKWB.S	6.2%

Risk Prone	
<i>highest return on the frontier</i>	
Expected return	17.63%
Risk	22.62%
Return / risk	0.779
Holdings	30
Weighted ESG	70.0
Top sector	Technology 30%
Top country	United States 60%
Largest positions	
ARGX.BR	20.0%
NVDA.OQ	20.0%
PWR.N	20.0%
GAW.L	10.8%
AVGO.OQ	3.0%

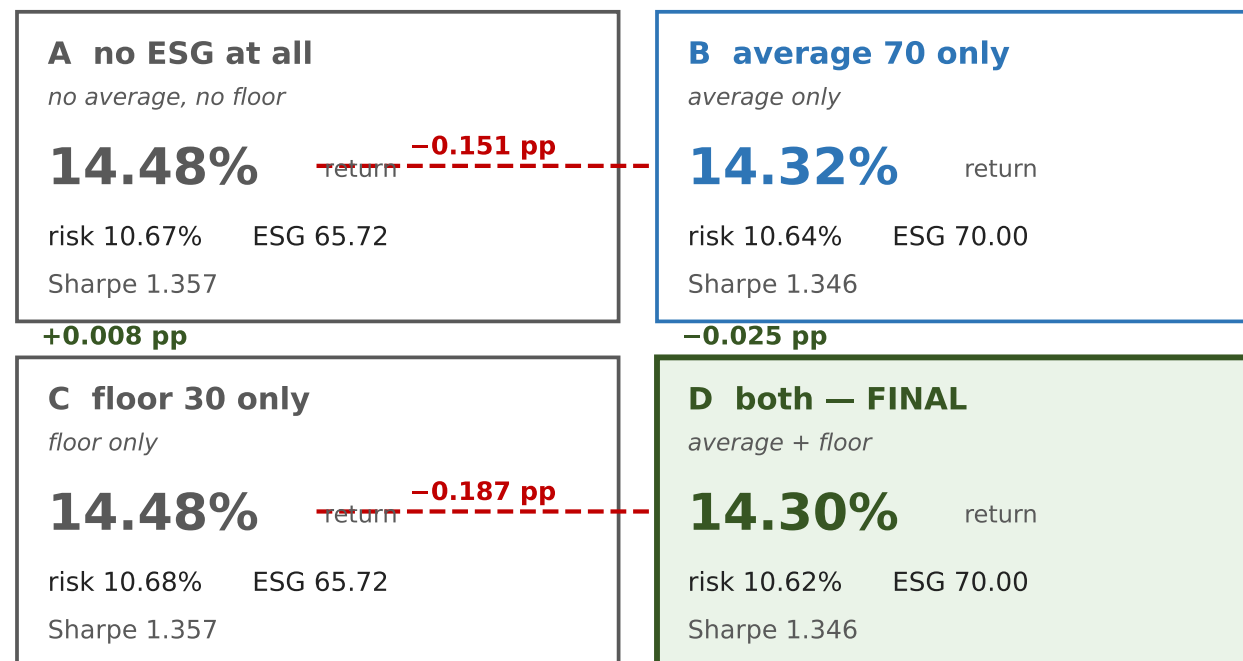
Read the risk-prone column with care.

"Maximise return" lands on a corner of the feasible set: 3 positions sit at the 20% cap and 23 of 30 at the 1% floor, so 60% of the budget is in three stocks. Its return/risk ratio of 0.78 is worse than the risk-averse book's 1.07 — it is a valid answer to the question asked, not a portfolio we would recommend.

ESG policy: four configurations, not two

The portfolio average and the per-stock floor are independent levers — each was toggled separately

Neutral profile shown in each cell



Which rule actually costs anything

Lever, cost in pp	Averse	Neutral	Prone
Floor alone A → C	+0.003	+0.008	+0.000
Average alone A → B	-0.025	-0.151	-0.270
Floor on top B → D	-0.045	-0.025	+0.000

- A and C move together, B and D move together. That is the whole finding. The per-stock floor changes nothing on its own — +0.008 pp at Neutral and exactly zero at Risk Prone, both inside solver tolerance.
- The average constraint carries the entire cost. -0.151 pp at Neutral, -0.270 pp at Risk Prone. Anything said about "what ESG costs" is a statement about this rule alone.
- So the floor is free insurance. Its job is not performance, it is closing the loophole where a weighted average of 70 hides a stock scored in the twenties — Berkshire Hathaway at 24.1 is the clearest example. A governance improvement at no measurable return cost.

Scenario coverage

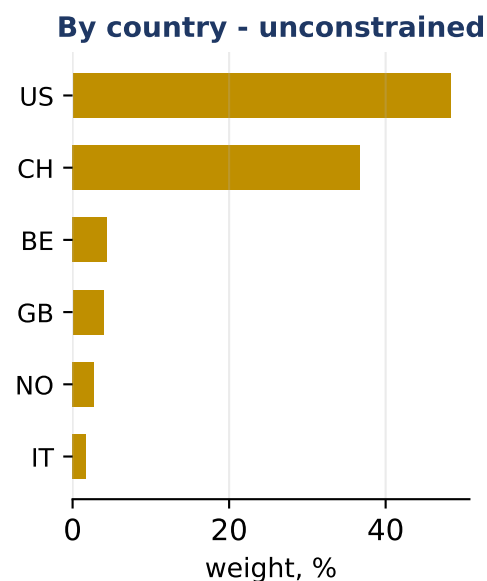
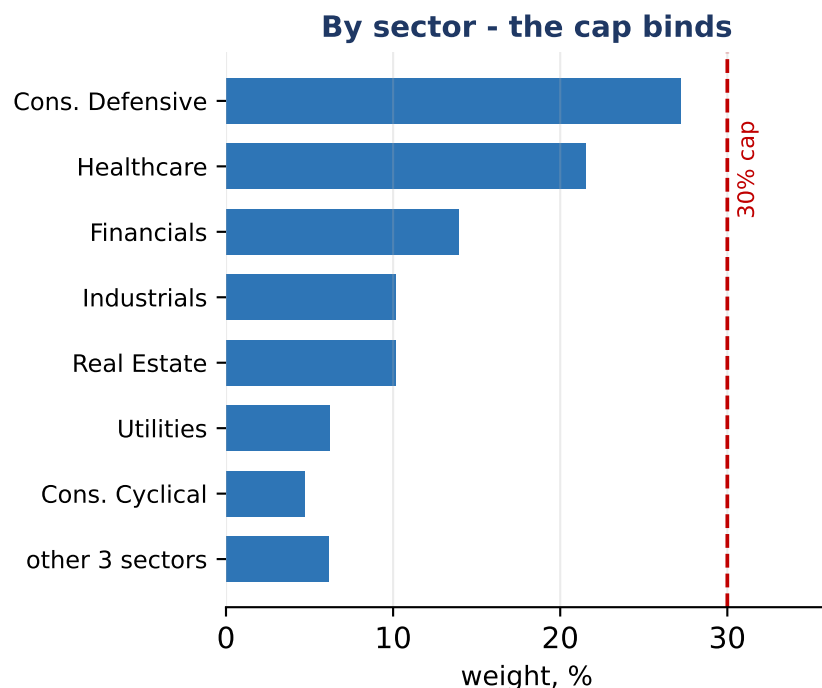
Every axis tested, and the two that were not, said plainly

Axis	What was varied	Result
1 Risk profile	min risk / max return-risk / max return, on every configuration below	the spine of the analysis: 9.87% at 9.26% risk up to 17.63% at 22.63%
2 ESG policy	the full 2x2: no ESG, floor only, average only, both	the average rule carries all the cost; the floor is free — see the previous slide
3 Controversial industry	Tier 1 (Rheinmetall) vs Tier 1 + ten defence primes	broad exclusion costs -0.004 pp at Neutral — a policy choice, not an economic one
4 ESG floor threshold	20 / 30 / 40	14.299% / 14.300% / 14.305% at Neutral — the answer does not depend on the number picked
5 Sector cap 30%	on vs off	off is worth +0.020 pp at Neutral, but raises Risk Prone volatility from 22.63% to 23.14% — the cap lowers risk where it binds
6 Estimation window	3-year rolling window, 31 quarterly re-estimations	answered by the walk-forward backtest, not by re-solving: it needs the data pipeline rerun
7 Out-of-sample behaviour	1/N benchmark, COVID and 2022 stress periods	done: Sharpe 0.835 vs 0.866, indistinguishable; wins in both crises, loses in rallies

Twenty-four solved scenarios: eight configurations x three risk profiles, plus the 31-rebalance backtest. Not run and worth saying so: a factor-level attribution of where the return comes from, and transaction costs inside the optimiser rather than applied afterwards.

What the recommended portfolio holds

Neutral profile - 30 positions, \$100M



Top holdings	Wt	USD	Ctry	ESG
WMT.OQ	12.9%	\$12.9M	US	77
GALE.S	10.2%	\$10.2M	CH	69
MOBN.S	10.1%	\$10.1M	CH	82
COST.OQ	8.2%	\$8.2M	US	75
BKWB.S	6.2%	\$6.2M	CH	58
PGR.N	5.9%	\$5.9M	US	69
LLY.N	4.7%	\$4.7M	US	67
RSG.N	3.5%	\$3.5M	US	76

The concentration the sector cap cannot see

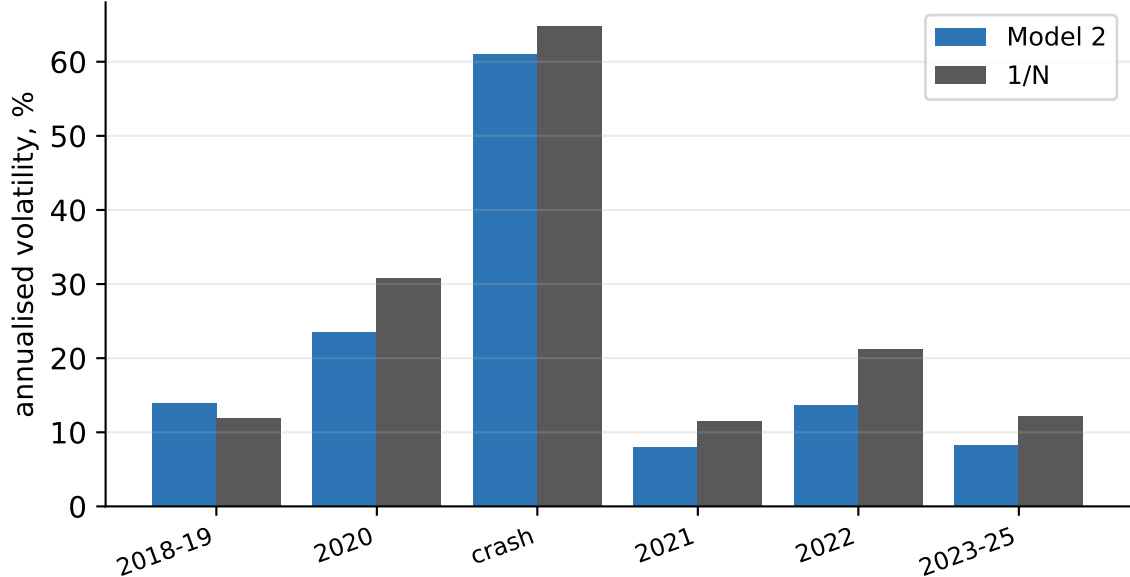
- A sector cap holds Consumer Defensive to exactly 30.00%, but nothing restrains country exposure - Switzerland reaches 47% of the risk-averse book, concentrated in cantonal banks and real estate.
- A 25% country cap held it to 25.0% for 0.156pp of return. It is not in the current model, and it is the most obvious next addition.

How the portfolio behaves across market regimes

Out-of-sample, from a walk-forward test: rolling 3-year window, quarterly rebalance, 31 rebalances over 7.7 years

Benchmark is an equal-weight (1/N) portfolio over the same eligible universe at each rebalance date, so the comparison isolates the optimiser rather than the choice of stocks.

Volatility, by regime



- The pattern is the point. It wins in the drawdowns and loses in the rallies - in the COVID crash it lost 3.7pp less than the benchmark, and in 2022 it fell less while running 36% lower volatility, giving that back in 2021 and 2023-25. That is the trade a low-volatility mandate explicitly buys, and the reason to present this as a risk-control product rather than a return-maximising one.
- One exception, stated rather than glossed: in the calm 2018-19 window it ran higher volatility than 1/N and earned +11pp for it. Volatility is lower in five of six sub-periods, not all six.

Period	Regime	Return Δ, pp	Vol change	Max DD
2018-19	calm market	+11.1	+17%	-11% vs -18%
2020 full	crash and rebound	-14.3	-24%	-34% vs -38%
the crash	the crash itself	+6.9	-6%	-34% vs -38%
2021	boom	-13.3	-30%	-7% vs -6%
2022	inflation shock	+0.6	-36%	-23% vs -27%
2023-25	AI rally	-7.2	-33%	-8% vs -13%

Max DD shown as model vs 1/N. Highlighted: the two periods the model wins.

Limitations, measured not guessed

Four of these are properties of the data and cannot be fixed with this dataset

Limitation	The measurement	Consequence for the reader
Risk is understated out of sample	predicted 8.78% at rebalances vs 13.21% realised, i.e. 50% understated	say "consistent with the estimation period", never "conservative"
The estimation window is a judgement	Lockheed is picked in 100% of 5-year bootstrap resamples and 50% of 10-year ones	we chose 10 years for a full cycle; the cost is reported, not hidden
ESG and sector data are a snapshot	one present-day value per stock, no history	applying ESG ≥ 70 at a 2018 rebalance uses 2025 information — look-ahead
The universe is survivorship-biased	today's STOXX 600 and S&P 500 constituents only	returns are optimistic for every strategy here, benchmark included
Country concentration is unconstrained	Switzerland reaches 47% of the risk-averse book	a 25% country cap costs 0.156pp — the obvious next addition
The 30-stock floor pads the book	11 of 30 positions sit exactly on the 1% minimum	a higher minimum weight would give a cleaner portfolio — worth raising with the client
Transaction costs are outside the model	turnover 21% per rebalance, ~83% a year	at 10bp that is 0.18pp of CAGR, so costs are not the binding issue

The first two we can act on. The third and fourth are properties of the dataset. The last three are choices open to the client.

Where this stands, and what is next

Settled

- Model formulation, solved to optimality in under a second on 1,087 stocks
- Data cleaning — every decision measured and reproducible from a script
- Both risk models compared: quadratic beats linear by 19.2% of risk
- Three portfolios, all seven constraints verified independently of the solver
- Walk-forward backtest, 7.7 years and 31 rebalances

Open

- Country cap. Switzerland at 47% is the one concentration nothing restrains. Measured cost 0.156pp.
- Dashboard. Not started, and the largest remaining piece of work.
- Six missing series. Three REITs among them — worth re-extracting.

The one slide to remember

- The optimiser reliably delivers what it optimises — 22% less volatility than equal weight, and smaller drawdowns in every crisis in the sample.
- It does not deliver a higher risk-adjusted return out of sample. That is a known result in the literature, not a defect in this implementation, and the cause — estimation error — we measured three independent ways.
- So this should be presented as a risk-control product, not a return-maximising one. That is what the constraints describe and what the backtest supports.

*Educational exercise. Not investment advice.
Past performance does not guarantee future results.*